a A recorded zero still contributes

Gene kept, count 0

Standardized value

$(0 - \mu_g)/\sigma_g$

Contribution to class score

$-w_{kg}\,\mu_g/\sigma_g$

Gene removed

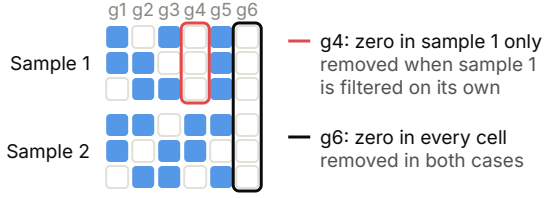
No value in the input

Contribution to class score

0

Removing the gene adds the same offset to a class score in every cell of the query: $\delta_k = w_{kg}\,\mu_g/\sigma_g$

b The query decides which genes are undetected



The same sample 1 cell is scored with g4 after pool filtering and without it after sample filtering.

c Inputs compared for the same 18,410 cells

Original counts
7 samples, 2 studies
18,410 cells

→ **Normalize, then exclude**
duplicate gene symbols
(10,000 counts per cell, log1p)

→ **Unfiltered**
all measured genes with
a unique gene symbol

Sample-filtered
removes genes undetected
in the sample

→ **Restored**
puts the removed genes
back as zeros

Pool-filtered
removes genes undetected
in the study pool

min_cells=3
removes genes detected
in fewer than 3 cells

Filtering contrast: unfiltered vs sample-filtered

Composition contrast: pool-filtered vs sample-filtered

Control: restored vs unfiltered, expected to match exactly. Secondary comparison: min_cells=3 vs unfiltered.